

LUCA GIANFORCARO

Biomechanics Engineer | Robotics and Machine Learning

Date of birth: 14/04/2000

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📍 Turin(TO), Italy

in luca-gianforcaro



EXPERIENCES/PROJECTS

Master's Thesis Project

Sant'Anna School of Advanced Studies

📅 Sep 2025-Ongoing

📍 The BioRobotics Institute,Pontedera

- Master's Thesis: "Arm-Hand Synergic Control for FES Actuation on Hybrid Exoskeleton"
- Conducted systematic literature review on control strategies for hybrid exoskeleton systems
- Data acquisition pipeline: implementation of UDP-based communication protocol for multi-device synchronization between the NESM upper-limb exoskeleton and the MANUS sensorized glove.
- Building of a multimodal dataset of upper limb kinematics collected during functional daily-life tasks.
- Data preprocessing pipeline: data synchronization, segmentation, labeling, and windowing for time-series deep learning models.
- Designed and trained a BiLSTM classifier for hand motion intent decoding from proximal arm kinematics; applied transfer learning to adapt the model from healthy to impaired subject data, addressing inter-domain distribution shift across heterogeneous user populations.
- Developed a generative model (VAE-based) for subject-invariant latent representation learning
- Model optimization and embedded deployment via TFLite quantization for real-time inference on resource-constrained hardware.

Master Degree Activities

Polytechnic University of Turin

📅 2023-Ongoing

📍 Turin-Italy

- Design artificial limbs and multibody modelling
- End-to-end workflow for artificial limb design: medical image segmentation, CAD modeling, FEM and multibody dynamics.
- Biomedical signals and systems analysis.
- Advanced signal processing, including filtering, noise reduction, and feature extraction, to analyze and interpret biomedical signals.
- MATLAB and Simulink for dynamic system analysis, modeling, and simulation
- Team-based Engineering Projects.
- Characterization of a dataset using machine learning algorithms,to predict the outcome of weaning patients from assisted mechanical ventilation.
- Design and presentation of a bioreactor aimed at the differentiation and maturation of skeletal myoblasts on polymeric scaffolds.
- Computational project focused on molecular dynamics simulations, utilizing software such as GROMACS and VMD.

Erasmus+ Project

NOVA School of Science and Technology

📅 February 2022-July 2022

📍 Lisbon-Portugal

- Erasmus+ Project in collaboration with NOVA School of Science and Technology (FTC NOVA). Focus on:
- Signal Theory, Biomechanics,Organic Chemistry,Language and culture portuguese course.

Study abroad programs

Foreign Universities

📅 2017-2018

📍 Miami-United States, Dublin-Ireland

- Keiser University -Miami- 2 weeks
- ATC school - Dublin- 2 weeks

EDUCATION

Master Degree in Biomedical Engineering

Polytechnic University of Turin

📅 2023-2026

📍 Turin,(TO)

Mark:110/110 with honor

Bachelor Degree in Biomedical Engineering

University of Palermo

📅 Sep 2019-Mar 2023

📍 Palermo,(PA)

Mark:110/110 with honor

Scientific High School

High School A.Sciascia

📅 Sep 2014-July 2019

📍 Canicatti,(AG)

Mark:96/100

LANGUAGES

- Italian: Native
- English: Certified C1 IELTS
- Portuguese: Certified B1

ACTIVITIES

• English Course C1 level

📅 2020-2021

📍 Wall Street English School

• Physics and Maths Tutor

📅 2019-2021

📍 GoStudent Platform

• Portuguese Course

📅 2021

📍 NOVA FCT of Lisbon, Portugal

SKILLS

Programming & ML

Python TensorFlow/Keras MATLAB/Simulink

Labview C++ (basic)

Robotics & Systems

ROS2 Embedded deployment

UDP real-time communication

Engineering Tools

Solidworks Mimics Patran-Nastran Gromacs

VMD Adams View Rhinoceros 3D Octave

Orcad Pspice

INTERESTS

Robotics Machine Learning AI Technology

Innovation Sports-Fitness Travelling

In compliance with the legislative Decree no. 196 dated 30/06/2003 and GDPR art. 3 (UE 2016/679), I hereby authorize you to use and process my personal details contained in this document.